

Local-AI Smart Traffic Lights with Emergency Vehicle Preemption: A Privacy-Aware Edge Architecture for Intelligent Traffic Management

Your Name

Department of Computer Science and Engineering

Your College / University

City, Country

your.email@example.com

Abstract—Traditional traffic signal systems commonly rely on fixed timing plans or detector-based actuation. Although these approaches are simple and reliable, they may respond slowly to rapidly changing traffic conditions. Recent research has investigated computer vision, artificial intelligence (AI), reinforcement learning, and edge computing as methods for improving traffic signal responsiveness.

This paper proposes a local-AI smart traffic-light architecture in which roadside cameras provide traffic information to an edge computing device located near the intersection. The edge device performs vehicle detection, traffic-state estimation, and emergency-vehicle recognition without requiring continuous transmission of raw video to a remote cloud server. An adaptive traffic signal controller uses the estimated traffic state to determine bounded signal-control actions, while a separate emergency-vehicle preemption module provides priority to verified emergency vehicles.

The proposed architecture combines computer vision, edge intelligence, adaptive traffic signal control, and emergency vehicle preemption. The cloud is used primarily for historical storage, analytics, model training, and system monitoring rather than for the hard real-time control loop. The paper also proposes a reproducible evaluation methodology using the Simulation of Urban MObility (SUMO). Evaluation metrics include average vehicle delay, queue length, travel time, emergency vehicle delay, emergency detection accuracy, false preemption rate, inference latency, communication volume, and recovery time following emergency preemption.

The objective is to provide a practical and safety-aware framework for developing intelligent traffic signals that can respond to changing traffic conditions while reducing dependence on continuous cloud connectivity.

Index Terms—Intelligent Transportation Systems, Edge AI, Smart Traffic Lights, Adaptive Traffic Signal Control, Computer Vision, Emergency Vehicle Preemption, Reinforcement Learning, Traffic Simulation

I. INTRODUCTION

Urban road intersections are important components of transportation networks because several conflicting traffic movements must share limited road capacity. Traditional traffic signals generally operate using pre-timed schedules or actuated control. Pre-timed systems use predefined green, yellow, and red intervals, while actuated systems can modify signal operation based on detector inputs [1].

Although conventional signal control is widely deployed, fixed timing may not respond efficiently when traffic demand changes significantly. For example, an intersection may have a long queue on one approach while the opposite approach has very few vehicles. A fixed-time controller may still allocate a predetermined amount of green time to both approaches.

Artificial intelligence and computer vision provide an alternative approach. Roadside cameras can be used to estimate vehicle counts, queue lengths, occupancy, movement patterns, and other traffic characteristics. Machine vision has been investigated for adaptive traffic-light control and traffic parameter estimation [2].

Deep reinforcement learning (DRL) has also become an active research area for traffic signal control. DRL models can learn relationships between traffic states and signal-control actions and can optimize objectives such as delay, queue length, and throughput [3].

Another important development is edge intelligence. Instead of sending all camera data to a centralized cloud server, computation can be performed near the source of the data. Edge intelligence has been investigated for intelligent transportation systems because it can reduce latency and network traffic while enabling local processing [4].

The student handout that motivated this research proposes a system based on street cameras, smart-pole computers, a city cloud server, and traffic lights. It also identifies automatic emergency-vehicle path clearance as an important research direction.

This paper expands that concept into a literature-based architecture that combines local computer vision, adaptive traffic signal control, edge computing, and emergency-vehicle preemption.

The main objectives of this research are:

- 1) To design a local-AI architecture for intelligent traffic signal control.
- 2) To use roadside computer vision for estimating traffic conditions.
- 3) To investigate edge computing as an alternative to cloud-only processing.

- 4) To integrate emergency-vehicle detection and signal preemption.
- 5) To propose measurable evaluation metrics for testing the proposed architecture.

II. RELATED WORK

A. Traditional Traffic Signal Control

Traditional traffic signals can be categorized into pre-timed and actuated systems. In pre-timed systems, signal phases operate according to predefined schedules. Actuated systems use detector information to modify signal operation according to traffic demand [1].

These systems are relatively deterministic and are therefore useful as baselines for evaluating intelligent traffic control.

However, conventional approaches can become inefficient when traffic demand changes rapidly or when unusual events occur.

B. Machine Vision for Traffic Control

Computer vision enables traffic signals to obtain information directly from roadside video.

Gorodokin *et al.* investigated adaptive traffic-light control based on machine vision and traffic-flow parameters [2]. Such systems can estimate the number of vehicles and other traffic characteristics and use this information for adaptive control.

Modern object-detection models such as YOLO have also demonstrated the ability to detect objects in real time. YOLOv4 was designed to provide a balance between detection accuracy and inference speed [5].

For traffic signal applications, however, object-detection accuracy alone is not sufficient. The complete system must also consider processing latency, camera placement, lighting, occlusion, communication delay, and controller response time.

C. Deep Reinforcement Learning

Deep reinforcement learning has been extensively studied for adaptive traffic signal control.

Rasheed *et al.* reviewed deep reinforcement learning approaches for traffic signal control and discussed how learning-based controllers can adapt their decisions according to traffic conditions [3].

A reinforcement-learning controller can be represented as an agent that observes the traffic state, selects an action, and receives a reward.

For example, the traffic state can be represented as:

$$s_t = [q_1, q_2, \dots, q_n, o_1, o_2, \dots, o_n, v_1, v_2, \dots, v_n] \quad (1)$$

where:

- q_i represents the queue length on approach i ,
- o_i represents road occupancy,
- v_i represents traffic speed.

The controller then selects an action a_t such as maintaining the current phase, extending the current green interval, or requesting a phase change.

D. Edge Intelligence

Cloud-based AI can provide substantial computational resources but requires communication between roadside devices and remote servers.

Edge intelligence moves computation closer to the data source. Gong *et al.* surveyed edge intelligence in intelligent transportation systems and discussed benefits including lower latency and reduced network requirements [4].

For traffic signals, this architecture is particularly useful because traffic decisions can be time-sensitive.

A local edge computer can receive camera frames, perform object detection, estimate traffic conditions, and provide control information without requiring every camera frame to be uploaded to a remote cloud.

E. Emergency Vehicle Preemption

Emergency vehicle signal preemption is an established traffic-control technique.

The Federal Highway Administration describes signal preemption as a transition from normal signal operation to a special operating mode for situations such as emergency-vehicle movement [6].

Emergency vehicle detection can be performed using different technologies, including optical systems, radio systems, acoustic detection, and other dedicated mechanisms.

Research has also investigated emergency-vehicle-centered traffic signal control to reduce emergency response delay while considering the effect on other traffic [7].

The proposed system therefore does not treat emergency preemption itself as a new concept. Instead, the research focuses on integrating emergency detection with local AI and edge computing.

III. PROBLEM STATEMENT

Existing traffic signal systems can perform effectively under predictable traffic conditions. However, traffic conditions may change rapidly because of congestion, accidents, road construction, weather, or emergency vehicles.

Cloud-based AI systems can process large quantities of data but may depend on network connectivity and introduce communication latency.

The research problem addressed by this paper is:

Can a local-AI traffic signal architecture reduce traffic delay and provide rapid emergency-vehicle preemption while minimizing dependence on continuous cloud connectivity?

The proposed research focuses on the integration of:

- 1) roadside computer vision,
- 2) edge computing,
- 3) adaptive traffic signal control,
- 4) emergency vehicle detection,
- 5) emergency signal preemption,
- 6) safety supervision.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed system consists of five primary components:

- 1) Street Cameras
- 2) Smart-Pole Edge Computer
- 3) Adaptive Traffic Signal Controller
- 4) Emergency Vehicle Preemption Module
- 5) Traffic Signal Controller

The cloud server is used for historical data storage, analytics, monitoring, and machine-learning model development.

The cloud is not placed inside the critical real-time signal-control path.

A. System Architecture Diagram

Figure 1 presents the proposed architecture.

B. Street Cameras

Street cameras continuously observe traffic approaching the intersection.

The cameras may provide:

- vehicle counts,
- vehicle classification,
- queue length,
- traffic density,
- vehicle speed,
- emergency vehicle detection.

The camera does not necessarily need to transmit the complete video stream to a remote server.

Instead, video can be processed locally.

C. Smart-Pole Edge Computer

The edge computer is responsible for local AI processing. Its primary tasks are:

- 1) receiving camera frames,
- 2) detecting vehicles,
- 3) estimating traffic states,
- 4) detecting emergency vehicles,
- 5) calculating confidence scores,
- 6) generating traffic-control requests.

Possible hardware platforms include embedded CPUs, GPUs, NPUs, or specialized AI accelerators.

D. Traffic-State Estimation

The traffic state can be represented as:

$$S_t = [Q_t, O_t, V_t, E_t] \quad (2)$$

where:

- Q_t = queue-length vector,
- O_t = occupancy vector,
- V_t = estimated speed vector,
- E_t = emergency vehicle confidence.

This state is provided to the adaptive controller.

TABLE I
PROPOSED SYSTEM FAILURE HANDLING

Failure Condition	Recommended Response
Camera failure	Fall back to conventional detector/control logic
AI inference failure	Ignore AI request and continue safe controller operation
Network failure	Continue local edge operation
Low emergency confidence	Do not trigger automatic preemption
Edge computer failure	Use certified fallback signal-control program

E. Adaptive Traffic Controller

The adaptive controller determines signal actions based on the current traffic state.

A generic reward function can be represented as:

$$R_t = -\alpha D_t - \beta Q_t + \gamma T_t - \delta P_t \quad (3)$$

where:

- D_t is traffic delay,
- Q_t is queue length,
- T_t is traffic throughput,
- P_t is a safety or constraint penalty.

The coefficients α , β , γ , and δ can be selected experimentally.

V. EMERGENCY VEHICLE PREEMPTION

Emergency vehicle preemption is treated as a high-priority supervisory function.

The system first detects a potential emergency vehicle and calculates a confidence value.

If the confidence is sufficiently high, the system estimates the vehicle's approach and expected arrival time.

A safe preemption request can then be generated.

A. Emergency Preemption Flowchart

Figure 2 illustrates the proposed emergency-vehicle decision process.

VI. SAFETY CONSIDERATIONS

AI should not bypass deterministic traffic-signal safety mechanisms.

The safety supervisor should verify that an AI-generated request satisfies minimum yellow times, clearance intervals, phase-conflict constraints, and other controller rules.

The system should also have fallback mechanisms.

This layered architecture prevents a single AI component from becoming the only mechanism responsible for traffic-signal safety.

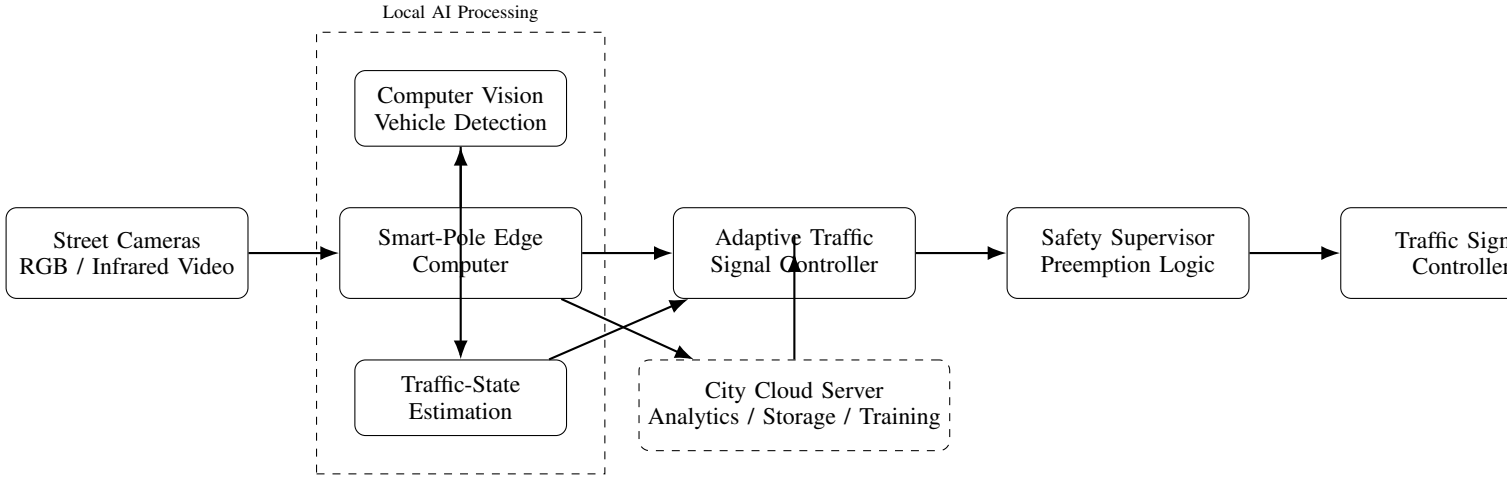


Fig. 1. Proposed local-AI smart traffic-light architecture. Camera data is processed locally at the edge while the cloud is primarily used for analytics, storage, and model development.

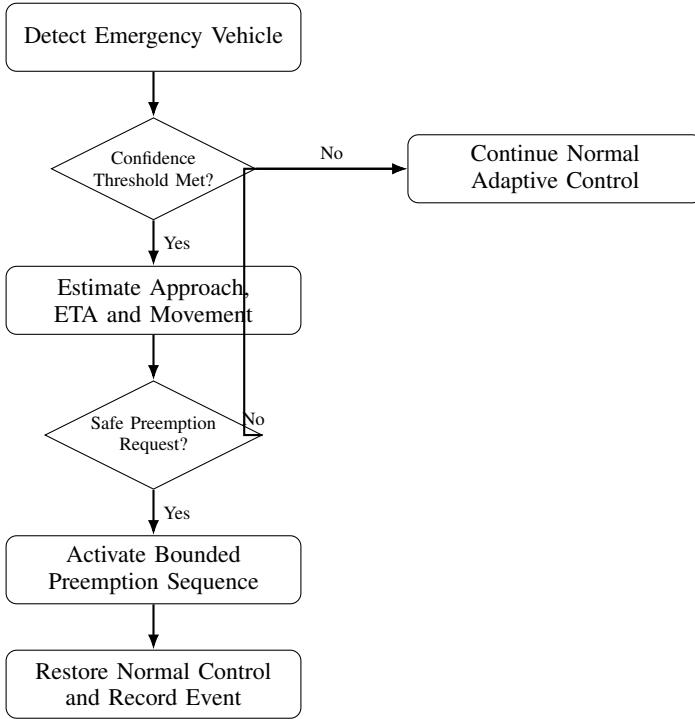


Fig. 2. Emergency vehicle detection and preemption decision flow.

VII. PRIVACY CONSIDERATIONS

Roadside cameras can capture large amounts of visual information.

A cloud-only architecture may require continuous transmission of video. The proposed architecture instead performs most real-time processing at the edge.

Only selected information can be transmitted to the cloud, such as:

- vehicle counts,
- traffic density,

TABLE II
ILLUSTRATIVE COMPARISON FROM THE SUPPLIED ASSIGNMENT HANDOUT. THESE VALUES SHOULD NOT BE INTERPRETED AS INDEPENDENTLY MEASURED RESULTS.

System	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30 s	None	\$200
Cloud AI	2 s	High	\$2,500
Local Smart Camera	0.5 s	Low	\$1,200

- queue length,
- signal timing information,
- system diagnostics.

This can reduce network requirements and reduce the amount of raw video leaving the intersection.

Edge intelligence has been studied as an approach for reducing network traffic and improving the responsiveness of intelligent transportation systems [4].

VIII. SYSTEM COMPARISON

The original assignment handout provides an illustrative comparison of three traffic-control architectures.

The values are reproduced here as assignment assumptions rather than experimental measurements.

These values should be experimentally validated before being presented as actual system performance.

The real latency of a local-AI system depends on hardware, camera resolution, neural-network architecture, inference framework, and communication with the traffic controller.

IX. RESEARCH METHODOLOGY

A reproducible evaluation can be performed using the Simulation of Urban MObility (SUMO), an open-source microscopic traffic simulator [8].

The proposed experiment consists of several traffic conditions.

A. Traffic Scenarios

- 1) Normal traffic
- 2) High traffic
- 3) Unequal traffic demand
- 4) Random traffic fluctuations
- 5) Emergency vehicle arrival
- 6) Emergency vehicle arrival during congestion

Each scenario should be repeated using multiple random seeds.

B. Baseline Systems

The following systems should be compared:

- 1) Fixed-time traffic signal control.
- 2) Conventional actuated traffic signal control.
- 3) Cloud-based AI traffic control.
- 4) Local edge-AI traffic control.
- 5) Local edge-AI control with emergency preemption.

C. Performance Metrics

The primary traffic performance metrics are:

$$\overline{D} = \frac{1}{N} \sum_{i=1}^N D_i \quad (4)$$

where $\overline{D}$ represents average vehicle delay.

Average queue length is:

$$\overline{Q} = \frac{1}{T} \sum_{t=1}^T Q_t \quad (5)$$

Average travel time is:

$$\overline{T} = \frac{1}{N} \sum_{i=1}^N T_i \quad (6)$$

Additional emergency-related metrics include:

- emergency vehicle delay,
- emergency detection accuracy,
- emergency detection latency,
- false preemption rate,
- route clearance time,
- recovery time after preemption.

For edge deployment, additional measurements should include:

- AI inference latency,
- end-to-end decision latency,
- CPU/GPU utilization,
- power consumption,
- network bandwidth.

X. EXPERIMENTAL PROCEDURE

The proposed experimental procedure is:

- 1) Create an intersection model in SUMO.
- 2) Configure realistic traffic demand.
- 3) Implement a fixed-time signal controller.
- 4) Implement an actuated controller.
- 5) Implement the local-AI controller.
- 6) Add emergency vehicle scenarios.
- 7) Add emergency preemption logic.
- 8) Execute multiple simulation runs.
- 9) Collect traffic performance measurements.
- 10) Analyze the results statistically.

The experiment should report both average values and variation across multiple runs.

XI. EXPECTED RESULTS

The proposed system is expected to provide a more responsive traffic control mechanism than a purely fixed-time system when traffic conditions change significantly.

The local-AI architecture is also expected to reduce dependence on continuous cloud communication because traffic-state estimation is performed at the edge.

For emergency situations, the preemption mechanism is intended to reduce the delay experienced by emergency vehicles.

However, these statements represent research hypotheses rather than experimentally established results. Actual performance must be measured using controlled experiments.

XII. LIMITATIONS

Several limitations must be considered.

First, computer vision performance depends on camera placement, weather, lighting, occlusion, image quality, and vehicle characteristics.

Second, an AI model trained for one intersection may not perform equally well at another intersection.

Third, simulation cannot perfectly reproduce real-world driver behavior.

Fourth, emergency vehicle detection must achieve a suitable balance between false negatives and false positives.

A false negative may cause an emergency vehicle to miss a preemption opportunity, while a false positive may unnecessarily disrupt normal traffic.

Finally, integrating an experimental AI system with a real traffic controller requires appropriate safety certification and engineering validation.

XIII. DISCUSSION

The literature demonstrates that the major components required by the proposed system already exist independently.

Machine vision can provide traffic-state information [2].

Deep reinforcement learning has been investigated for adaptive traffic signal control [3].

Edge intelligence has been investigated for intelligent transportation systems [4].

Emergency vehicle signal preemption is an established transportation technology [6].

The research opportunity therefore lies in integrating these technologies into a practical architecture.

One important design principle is separating ordinary traffic optimization from emergency response.

A controller that optimizes only average traffic delay may not provide the best emergency response.

Therefore, the proposed architecture uses an emergency preemption layer above the ordinary adaptive controller.

Another important consideration is reliability.

The traffic signal must remain operational even if the camera, AI model, edge computer, or cloud connection fails.

Consequently, conventional controller functionality should remain available as a fallback.

XIV. FUTURE WORK

Future research can extend the proposed system in several directions.

A. Multimodal Emergency Detection

Vision-based detection can be combined with:

- siren detection,
- authenticated vehicle-to-infrastructure messages,
- dedicated optical transmitters,
- radio-based emergency detection.

Combining multiple sources may improve detection reliability.

B. Multi-Intersection Coordination

Future systems can coordinate multiple traffic signals along an emergency vehicle route.

Instead of preempting only one intersection, the system could estimate the vehicle's route and prepare a sequence of intersections.

C. Hardware Benchmarking

Future experiments should compare different edge-computing platforms.

Measurements should include:

- frames per second,
- inference latency,
- energy consumption,
- model accuracy,
- hardware cost.

D. Privacy-Preserving AI

Future implementations should investigate methods that minimize raw video storage and transmission.

Only the traffic information required for traffic control and analytics should be retained whenever possible.

XV. CONCLUSION

This paper presented a local-AI architecture for intelligent traffic signal control with emergency vehicle preemption.

The proposed architecture uses roadside cameras to observe traffic, performs AI processing on a local edge computer, estimates traffic states, and provides information to an adaptive signal controller.

An emergency vehicle detection and preemption layer is added to provide priority during emergency situations.

The cloud is primarily used for data storage, analytics, monitoring, and model development rather than for the immediate signal-control loop.

The literature supports the use of machine vision for traffic information, learning-based traffic signal control, edge intelligence, and emergency vehicle preemption.

The proposed research framework can be evaluated using SUMO and controlled experiments involving normal, congested, variable, and emergency traffic conditions.

Important evaluation metrics include traffic delay, queue length, travel time, emergency vehicle delay, detection accuracy, false preemption rate, processing latency, bandwidth consumption, and recovery time.

The architecture is intended as a research framework rather than a claim of completed field performance. Future work should validate the system through simulation, hardware testing, and eventually controlled real-world experiments.

ACKNOWLEDGMENT

The initial research problem and architecture were derived from the student handout supplied for this assignment. External academic literature and transportation guidance were used to expand and verify the research framework.

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